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Faculty of Exact Sciences and Computer Science
Mathematics – L3 Applied Mathematics

Family Name:

First Name:

Year : 2026-2027
Date : 03/11/2026
Duration: 40 minutes

Test in Information System and Databases

Exercise 1: (05 points) **1 point for each correct answer**

Answer each of the following questions by checking the right box:

Q1- To share my printer with my coworkers, we need to be at lease in the same:

☐ Office	■ Intranet	☐ Internet	☐ Other

Q2- The Administrator of a database is:

☐ The Creator of the database	☐ An input device	☐ A regular user of the database	■ Other

Q3- Matlab is:

☐ An input device	☐ A utility software	☐ A productivity software	■ Other

Q4- Ubuntu is:

■ An operating system	☐ A utility software	☐ an application software	☐ Other

Q5- A Laser printer is:

☐ An input device	■ An output device	☐ an input-output device	☐ Other

Exercise 2: (6 points)

Given the following table **Customer** table in the relational database: **Store.** Answer the following questions:

Customer(customer_id, first_name, last_name, email)

customer_id	first_name	last_name	email
1	JAMES	MILLER	james.miller@sakilacustomer.org
2	EMILY	DAVIS	emily.davis@sakilacustomer.org
3	MICHAEL	SMITH	michael.smith@sakilacustomer.org
4	SARAH	JOHNSON	sarah.johnson@sakilacustomer.org
5	DAVID	JONES	david.jones@sakilacustomer.org
6	JESSICA	BROWN	jessica.brown@sakilacustomer.org
7	ROBERT	WILSON	robert.wilson@sakilacustomer.org

Q1- What is the: **(1 point)** **0.25 point for each correct answer**

- degree of the table: **4** , the number of records: **7** ,the number of rows: **7**, the number of attributes: **4**

Q2- What is the data type of **(2 points)** : **1 point for each correct answer**

- customier_id: **number** , first_name: **text**

Q3- What is the domain of values of **(2 points)**: **1 point for each correct answer**

- customer_id: **Strictly positive integers**
- email: F**irst name followed by a dot, followed by the last name, followed by @sakilacustomer.org**

Q4- We want that each customer can have only 2 phone numbers to contact him. What are the modifications to apply? **(1 point)**

Customer(customer_id, first_name, last_name, email,phone1, phone2)

Exercise 3: (04 points) **1 point for each correct answer**

Fill in the blank:

My team and I gathered today to discuss the development of the application we are creating for our new customer: a book store owner. The project is small (not big), so we are not going to use the **SDLC** development method. We will not be using any of the Agile methodologies. Nevertheless, we will start by specifying all the requirements of that application. So, we will be using the **RAD** development methodology. We will after organize **interactive** work sessions with a user's representative, for the design and the **construction** of the application.